CS-747: Deep Learning for Computer Vision

Course Project — Guidelines & Deliverables

Important — Topic Allocation Across Clusters

There are 6 clusters (5 topics each, 30 topics total) and 26 students forming the class. Topic selection must be balanced across clusters. Ensure exactly 4 topics are taken up from every cluster before any cluster is allowed a 5th.

First-come-first-served

Deliverable Pipeline:

Every project, regardless of topic, follows the same end-to-end pipeline. Each stage builds on the previous one, so treat this as a sequence rather than a checklist to complete in any order:

1. Problem Formulation —- define precisely what you are solving and why it is non-trivial.
2. Related Work — position your problem against existing literature, including the assigned reference paper.
3. Proposed Methodology — describe the architecture/pipeline/algorithm you will implement.
4. Hyperparameter Choices — justify the concrete training configuration you used.
5. Evaluation Metrics — define how success is measured, quantitatively.
6. Visualization of Results & Comparison with SOTA — show results visually and benchmark against the state of the art.
7. Ablation Studies — isolate which design choices actually matter.
8. Summarization of Findings — close the loop: what did you learn, what worked, what didn't.

The final report must integrate all eight stages into a single coherent IEEE-format paper — not eight disconnected sections stapled together (see format requirement below).

1. Problem Formulation

- State the task as a precise input → output mapping (e.g., "given an RGB image, predict a per-pixel class label").
- Explain why the problem is hard: what specifically makes it non-trivial (scale, domain shift, label scarcity, real-time constraints, etc.), tied to your assigned topic.
- Define the scope explicitly — what is in-scope vs. out-of-scope for your project (e.g., which dataset subset, which classes, which constraints you will and won't handle).
- State your hypothesis or research question in one or two sentences before diving into methodology.

2. Related Work

- Cover the assigned reference paper in depth — summarize its core contribution, method, and results in your own words (no verbatim copying).
- Include at least 4–6 additional related papers, organized thematically rather than as a flat list.
- For each related work, state what it does AND what its limitation is relative to your problem — this is what motivates your approach.
- Use IEEE-style in-text citation numbering, e.g., [1], [2], consistent with the References section.

3. Proposed Methodology

- Provide a clear architecture/pipeline diagram (block diagram or flowchart) of your approach.
- Describe each module/component and how data flows through the system, end to end.

- Explicitly state what is your own implementation vs. what is adapted/reused from existing code or pretrained models.
- If you deviate from or extend the reference paper's method, explain the modification and its motivation.

4. Hyperparameter Choices

- Report all key hyperparameters: learning rate, batch size, optimizer, number of epochs, loss function, regularization, augmentation, etc.
- Justify at least the 2–3 most impactful choices — don't just list values, explain why those values were chosen (grid search, prior work, compute constraints).
- Report hardware/compute budget used, since it affects what hyperparameter search was feasible.

5. Evaluation Metrics

- Choose metrics standard to your sub-area (e.g., accuracy/AUROC/FPR95 for classification & OOD, mAP for detection, mIoU/Dice for segmentation, FID for generative models) — accuracy alone is not sufficient where a domain-specific metric exists.
- Report metrics on a proper held-out test/validation split, never on training data.
- Where relevant, report both an aggregate number and a breakdown (e.g., per-class, per-size-bucket, base vs. novel class).

6. Visualization of Results & Comparison with State of the Art

- Include qualitative visual outputs (sample predictions, attention/saliency maps, generated images, segmentation overlays, etc.) — not just tables of numbers.
- Include at least one plot showing a trade-off relevant to your topic (e.g., accuracy vs. latency, FID vs. training time, precision–recall curve).
- Build a comparison table placing your result alongside the numbers reported by the state-of-the-art method(s) from your Related Work section, on the same dataset/protocol where possible.
- Be explicit and honest about any conditions under which your comparison isn't perfectly apples-to-apples (e.g., fewer training epochs due to compute limits).

7. Ablation Studies

- Identify at least 2 design choices in your method and test the system with each one removed or altered, keeping everything else fixed.
- Present ablation results as a table so the effect of each component is directly comparable.
- Interpret the results — state what each ablation tells you about why your method works (or doesn't).

8. Summarization of Findings

- Summarize what was achieved relative to the original problem formulation and hypothesis.
- State the most important 2–3 takeaways in plain language, without re-listing every number already in the results section.
- Honestly discuss limitations and what you would try next given more time/compute.

Report Format Requirement

The final project report must be written and formatted as an IEEE conference-style paper (two-column format), not a general Word report. This includes the standard IEEE structure: Title, Author(s), Abstract, Keywords, Roman-numeral Sections (Introduction, Related Work, Methodology, Experiments/Results, Ablation, Conclusion), and a numbered References section in IEEE citation style.

Use the official IEEE conference template (Word or LaTeX) as your starting point rather than building the two-column layout from scratch:

- *IEEE Manuscript Templates for Conference Proceedings — official Word/LaTeX templates: https://www.ieee.org/conferences/publishing/templates.html*

Reference

[1] IEEE, “IEEE Manuscript Templates for Conference Proceedings,” IEEE Author Center, 2024. [Online]. Available: https://www.ieee.org/conferences/publishing/templates.html